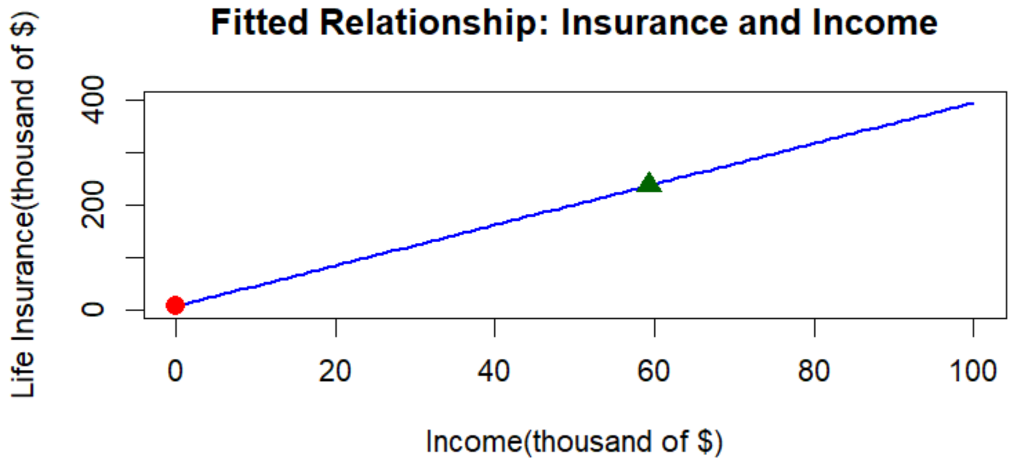
HW2 314707001 彭甫詔

3.18

(a.) sample mean of the amount of insurance: 236.939



(b.)

- The point estimate is 3.880 thousand dollars, which equals $3,880.
- The critical t-value at a 95% confidence level is approximately 2.101.
- The 95% interval estimate is [3.645, 4.115] thousand dollars, or [$3,645, $4,115].

Based on our sample of 20 households, our best estimate is that for every additional $1,000 a household earns in income, the average amount of life insurance they purchase increases by $3,880. While this single number is our best estimate, we have calculated a margin of error. We can say with 95% confidence that the true average increase across our entire market falls somewhere between $3,645 and $4,115 for each additional $1,000 in household income.

(c.)

- The point estimate is 394.855 thousand dollars
- The critical t-value at a 99% confidence level is approximately 2.878.
- The 99% interval estimate is [378.8936 , 410.8164] thousand dollars

Based on the results, we can see with 99% confidence that the expected average amount of life insurance held by households with an income of $100,000 is between $378,897 and $410,813.

(d.)

- Null Hypothesis (H_0): β_2= 5
- Test Statistic(t) = -10.0
- Rejection Region: Reject H_0 if t $\leq$ -2.101 or t $\geq$ 2.101

The calculated test statistic is t = -10.0. For a two-tailed test at the 5% level of significance with 18 degrees of freedom, the rejection region is t $\leq$ -2.101 or t $\geq$ 2.101. Since -10.0 falls deeply within the rejection region, we strongly reject the null hypothesis (H_0: β_2 = 5). Therefore, the data do not support the board member's claim. The actual average increase in life insurance for every $1000 increase in income is statistically significantly different from $5000.

(e.)

- Null Hypothesis (H_0): β_2= 1
- Test Statistics(t): 25.71429
- Rejection Region: Reject H_0 if t $\geq$ 2.552

Since our calculated test statistic (t = 25.714) is strictly greater than the critical value (2.552), it falls well within the rejection region. We strongly reject the null hypothesis at the 1% level of significance. We conclude that there is overwhelming statistical evidence to support the claim that as income increases, the amount of life insurance held increases by a significantly larger amount than the increase in income itself (i.e., the slope is definitively larger than one).

3.23

(a.)

- Null Hypothesis (H_0): $\beta_2 \leq \frac{13}{40}$
- Test Statistics(t): -25.22055
- Rejection Region: Reject H_0 if t $\geq$ 1.647919
- p-value $\approx$ 1

Since our calculated test statistic (t = -25.22055) does not fall into the rejection region, and the p-value (1.000) is much greater than the 0.05 significance level, we fail to reject the null hypothesis. We conclude that there is no statistical evidence to support the alternative hypothesis. The data indicate that the marginal effect on the expected house price of increasing the size of a 2000 square foot house by 100 square feet is indeed less than or equal to 13,000.

(b.)

- Null Hypothesis (H_0): $\beta_2 \leq \frac{13}{81}$
- Test Statistics(t): 4.571173
- Rejection Region: Reject H_0 if t $\geq$ 1.647919
- p-value $\approx$ 3.062619e-06

Since our calculated test statistic (t = 4.571173) fall into the rejection region, and the p-value is much smaller than the 0.05 significance level, we reject the null hypothesis. We conclude that there is a statistical evidence to support the alternative hypothesis. The data indicate that the marginal effect on the expected house price of increasing the size of a 4000 square foot house by 100 square feet is indeed greater to 13,000.

(c.)

- The point estimate is 167.3735 (thousand dollars)
- The critical t-value at a 99% confidence level is approximately 1.964739
- The 99% interval estimate is [158.0481 , 176.6988]thousand dollars

We estimate that the average selling price of houses with 2000 square feet of living area is about 167,373 dollars. A 95% confidence interval means that, based on this sample and model, we are 95% confident that the true average selling price of all such houses lies between158,048 and 176,699 dollars

(d.)

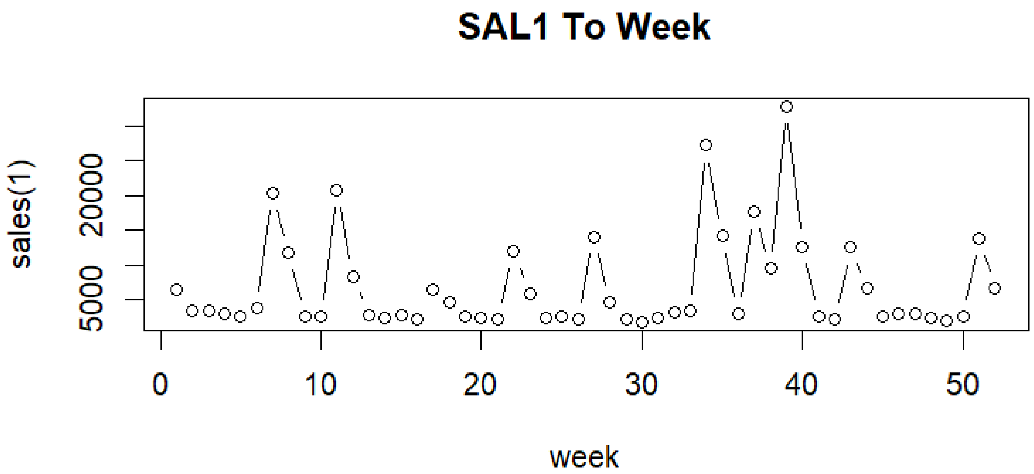
The sample mean selling price of houses with SQFT=20 is compatible with the result in part (c), since it lies within the 95% confidence interval for the expected price. This suggests that the regression model gives a reasonable estimate of the average selling price for 2000-square-foot houses.

3.31

(a.)

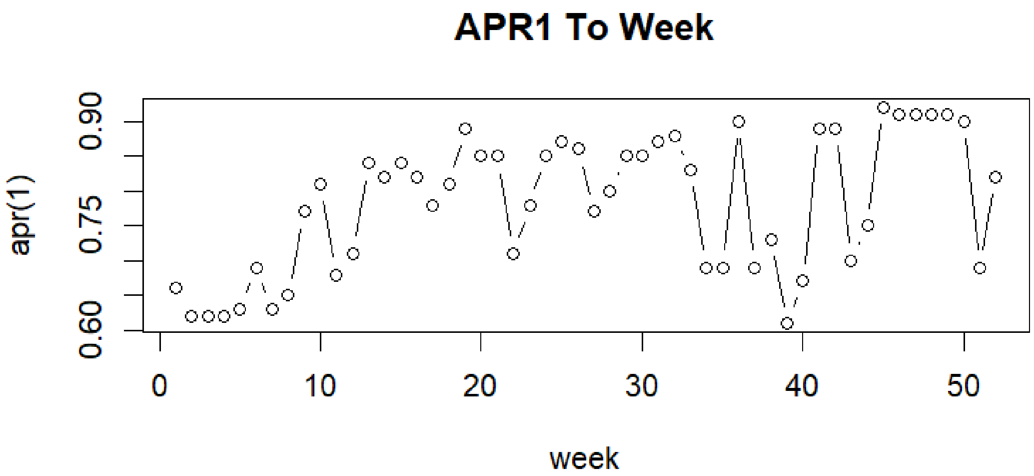
Summary Statistic for SAL1

Mean	Min	Max	SD
6718.712	1762.000	32820.000	6915.083



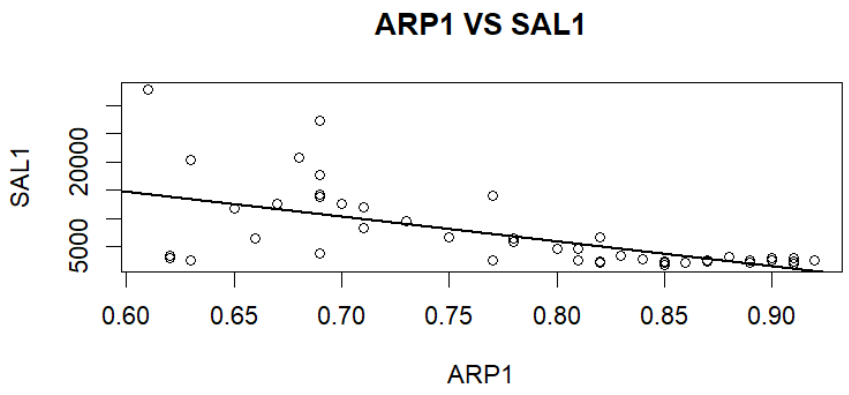
Summary Statistic for APR1

Mean	Min	Max	SD
0.78250000	0.61000000	0.92000000	0.09803711



Overall, sales show **considerable** week-to-week variation, while , while price shows comparatively little variation.

(b.)



The scatter plot of SAL1 against APR1 appears to show an inverse relationship: weeks with higher prices tend to have lower sales. This is consistent with economic intuition and the law of demand. When the price of brand no. 1 tuna increases, consumers tend to buy fewer units, so sales fall. Therefore, this is the relationship I would expect.

(c.)

- The point estimate is -434.4473
- The critical t-value at a 95% confidence level is approximately 2.009.
- The 95% interval estimate is [-592.29 , -276.6]

(d.)

- The point estimate is 10302.9
- The standard error of expected value is 1001.231
- The critical t-value at a 90% confidence level is approximately 1.675905.
- The 90% interval estimate is [8624.934 , 11980.87]

(e.)

- The estimated elasticity at the means is -5.0598
- The standard error of elasticity is 0.9152
- The 90% interval estimate is [-6.8981 , -3.2215]

The estimated price elasticity is negative and its absolute value is greater than one, indicating that sales are fairly elastic with respect to price. This is economically plausible because canned tuna likely has close substitutes, so consumers may respond strongly to price changes.

(f.)

Let $\alpha = \beta_2 * \frac{\overline{price1}}{\overline{sales1}}$

- Null Hypothesis (H_0): $\alpha = -3$
- Test Statistics(t): -2.2506
- Rejection Region: Reject H_0 if $|t| > 1.6759$
- p-value $\approx$ 0.0288

At the 10% significance level, we reject the null hypothesis that the elasticity equals −3. The data suggest that the true elasticity is different from −3.